



Per-Stream Terminal Valence Closure:
A Boundary Hypothesis and
Framework for Empirical Testing

Stephen Takowsky

Publishing Details

Per-Stream Terminal Valence Closure: A Boundary Hypothesis and Framework for Empirical Testing

Copyright © 2026 Law of Fairness LLC. All rights reserved.

Personal, educational, and non-commercial use of this complimentary digital edition is welcome, including scholarly discussion, review, quotation, and public commentary on social media and elsewhere. This edition may not be sold or distributed for commercial gain without prior written permission.

Published in the United States of America.

Publisher: Law of Fairness LLC

Author: Stephen Takowsky

Illustrations: Diego Santos

For permissions, corrections, publication updates, and future edition information: team@lawoffairness.com

Copies of this publication should be checked against the most current authorized edition.

Abstract

Per-Stream Terminal Valence Closure (PTVC) presents a precise, falsifiable candidate boundary hypothesis at the intersection of affective science, consciousness research, measurement theory, stochastic processes, and endpoint modeling. On its primary finite signed-measure branch, it asks whether each separately routed maximal qualifying conscious stream carries exactly equal positive and negative Jordan masses across its complete strict-pre-endpoint domain, so that the terminal signed residual is zero. The equality is intrastream and nontransferable. This compact boundary condition is compatible with the full diversity of lived experience: lives may differ in gross valence variation, duration, intensity, order, agency, and structure.

$$Z_i^* := \nu_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -).$$

$$\text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0.$$

The boundary condition becomes scientifically testable when joined to a prospectively fixed architecture for representation, sign, origin, scalarity, exposure, bearer, ownership and routing, domain, endpoint, observation and missingness, measured-to-latent recovery, reference class, and modality. Within that architecture, endpoint timing supplies a particularly informative structural discriminator for PTVC. On a prospectively declared positive-density endpoint branch under the at-risk law P_i^0 , suppose the endpoint probe is independent of the sigma-field generated by a jointly measurable pre-endpoint ledger path and has a conditional density strictly positive Lebesgue-almost-everywhere on a deterministic interval. If exact closure holds at the probe P_i^0 -almost surely, then, for P_i^0 -almost every eligible path, the ledger is zero Lebesgue-almost-everywhere on that interval. State-coupled timing, first hitting or absorption, endpoint atoms, singular carriers, routing change, continued experience, and endpoint-inclusive value define distinct, testable branches.

PTVC supports a research program structured for preregistration. Representation and origin validity, generator recovery, and observation

coverage form its foundation; random-cut, signed-dose, horizon, reserve, completion-potential, and comparative-route studies probe declared models. Informative indirect tests can precede exact complete-stream reconstruction; their bearing on PTVC depends on the tested implication and its independently justified bridges. On the declared action-first route, fixed-policy probability-one completion requires probability-one policy mass on actions whose complete frozen continuations close with probability one. Topological support, motivation, endpoint support, and all-intervention scope remain distinct. Neighboring literatures supply instruments, comparators, data, and safe perturbations. Differential route support requires frozen predictions to transport across held-out modules and outperform adequate equal-budget rivals; direct target adjudication uses its own branch-matched rules. Finite-resolution compatibility, an atom at zero, probability-one closure, and set-universal closure remain distinct. Complete-stream inference requires complete-domain accounting through observations or justified bounds, with validated endpoint and measured-to-latent bridges. The contribution joins a consequential conjecture to measurable signatures, discriminating tests, and adjudication of a possible terminal organizing principle of conscious experience.

Keywords: PTVC; Law of Fairness; valence; signed measures; terminal boundary condition; latent-variable identification; consciousness; endpoint processes; completion potential; affective science; partial identification; model recovery

Contents

Abstract	i
1. Scientific question, scope, and research opportunity	1
A boundary hypothesis with a precise domain	2
What makes the claim scientific	2
Scientific foundation and open empirical question.....	3
Contribution and novelty	3
2. Formal object and primary finite-measure claim	4
A latent ledger requires an exposure model.....	5
Bipolar and generalized branches	6
3. Quantifiers, ownership, routing, and falsification	7
Stream-level adjudication of the universal claim.....	8
Bearer and routing rules are causal, not clerical	8
Modal claims form a partly ordered family	8
Target and route are separate axes	9
Decision status remains typed	9
4. Representation, neutral origin, and delayed entry	10
Neutral origin must be calibrated independently	11
Study entry and stream origin are distinct	11
Factorization, scalarity, and gauge freedom	12
5. Observation, exactness, and the inference ladder	13
Target geometry and exactness levels	14
Inference must preserve the object type	15
Equivalence requires affirmative evidence.....	15
Exact atoms require stable inverse recovery.....	15
6. Random endpoints: a sharp structural discriminator	16
Independent positive-density endpoint proposition	17
What the discriminator reveals	17
Empirical use	17
What the proposition establishes and which branches it opens	18
Approximate dependence yields a graded pressure test	18
7. Reachability, endpoint support, and solvency	19
A residual must be cancelable by an admissible future	20
Endpoint support localizes the required structure.....	20
Endpoint adjudication is versioned evidence.....	21
Endpoint branches are empirical alternatives	21
8. Optional mechanism routes and path-class constraints	22
Finite variation restricts exact diffusion bridges.....	23
Reserve and endpoint require a tested bridge	24
Mechanism-localization discipline	24
Action-mediated route	24
Exact-differential route.....	25

9. Completion potential and the transport fingerprint	26
Completion potential	27
The overidentification fingerprint.....	28
Conjunctive differential signature	28
Random-cut complementarity	28
Randomized remainder and delayed-treatment localization.....	30
10. Rival tournament under equal information budgets.....	32
11. Existing evidence and the scientific opportunity	35
Established neighboring results	36
Plausibility signals and evidential roles.....	36
How independent convergence builds confidence.....	37
Endpoint, horizon, and consciousness literatures	37
12. Measurement, missingness, and endpoint validity	38
Representation before accumulation.....	39
Observation coverage completes the estimand	39
Missing not at random is a central rival.....	39
No-lookahead and reactivity.....	40
Replicate channels and recoverability	40
13. Confirmatory decision structure and model recovery	41
Recovery before interpretation	43
Negative controls and open-set alternatives	44
Unresolved cases remain visible.....	44
14. Safe nonterminal program: calibration before terminal claims.....	45
Stage 0 — protocol fixation, public preregistration, and rigorous design	46
Stage 1 — representation and local complementarity	46
Stage 2 — frozen completion-geometry transport	46
Promotion criterion.....	47
Immediate analyses in existing data	47
Three high-value experiments	48
Comparative route and future-regime tests.....	48
15. Endpoint-aligned extension and complete-stream adjudication	51
Observational bridge studies establish the path to terminal tests.....	52
Prospective endpoint architecture	52
Primary estimands	52
Evidence requirements for complete-stream adjudication.....	53
Endpoint surprise and promotion.....	53
16. Exactness, atoms, and symmetric error control	54
Boundary-aware inference.....	55
Nested-resolution discipline	56
Symmetric error-control budget	56
17. Falsification, safeguards, ethics, and research agenda.....	57
Results that would narrow or reject a protocol-fixed PTVC claim.....	58
Known challenges and their research solutions	58
Ethical boundary.....	58

What should materially change belief59

Near-term prerequisite and differential predictions59

Early publishable milestones60

Cumulative decision bookkeeping.....60

Promotion, retirement, and open challenge60

Ordered research agenda61

18. Evaluation and reproducibility standard62

Modular, noncompensatory evaluation.....63

Minimum reproducibility capsule.....63

Independent audit and discrepancy handling.....63

Promotion, retirement, and publication outcomes64

Declarations64

References65

About the Author72

1. Scientific question, scope, and research opportunity

The scientific question is sharply defined: after independently fixing a qualifying bearer, exhaustive ownership and routing, a complete strict-pre-endpoint truth domain, prospectively fixed valence representation, experienced-exposure rule and clock, neutral origin, and truth-level endpoint, are the positive and negative Jordan masses finite and equal for every separately routed maximal qualifying stream in the declared class?

Prospective-specification convention. Protocol-fixed means written into the study protocol before outcome inspection. Prospectively specified means fixed before outcome inspection without asserting a public record. Publicly preregistered means deposited before outcome inspection in a timestamped external registry with a persistent identifier; cite that identifier when making the claim. Without it, use protocol-fixed or prospectively specified.

A boundary hypothesis with a precise domain

PTVC concerns one cumulative invariant while welfare, opportunity, agency, duration, remembered utility, social outcomes, and gross experience remain distinct dimensions. A stream containing large positive and negative magnitudes may satisfy the same terminal residual condition as a stream containing small magnitudes. This separation preserves the independent importance of rights, justice, suffering prevention, dignity, and practical welfare. Every application of the hypothesis must uphold treatment, protection from harm, and active concern for suffering.

Interpretive scope. PTVC concerns the terminal within-stream relation and leaves path monotonicity and the number of intermediate zero crossings open. Equality of aggregate positive and negative masses does not imply one-to-one event pairing. PTVC does not imply that a bearer knows or computes a ledger, that every urge is corrective, or that psychological motivation is the mechanism. It asserts no debt, punishment, desert, payment, causal payback, or personal guidance.

What makes the claim scientific

Scientific force comes from prospectively fixed gates. The architecture fixes representation, bearer, domain, endpoint, observation, recovery, reference class, and modality before analysis. Route-specific claims

compare frozen predictions with equally resourced representation, endpoint-alignment, missingness, instrument-floor, and conditioned-model rivals. Direct target adjudication instead follows its residual and quantifier rules; a verified target-level counterexample need not identify its mechanism. Each passed gate advances only its linked claim.

Scientific foundation and open empirical question

PTVC is mathematically specified and empirically adjudicable under explicit measurement, bridge, and identification assumptions. Existing work on affective measurement, adaptation, opponent regulation, computational mood, consciousness, terminal trajectories, and missing data supplies candidate instruments, models, and validation methods. This article states a precise target, derives conditional implications, and supplies a preregistration-ready research framework; it reports no new empirical test establishing PTVC. Complete-stream adjudication remains a culminating objective, while a prospectively specified route or consequence restriction can be tested earlier when its implication and observation bridge are independently justified.

Interpretive context. Ideas of balance, restoration, consequence, and completion can generate comparative hypotheses across cultural narratives and lived experience. PTVC-specific scientific weight arises when prospectively specified terminal and cross-module predictions outperform cultural transmission, memory, comfort, cooperation, just-world belief, and other serious rivals.

Contribution and novelty

The article's distinctive contribution is the typed assembly of established signed-measure, stochastic-process, reachability, conditioning, identification, and model-recovery tools around the PTVC target, together with the branch-specific consequences derived from that assembly. The result is a novel, integrated research architecture: an exact intrastream terminal condition linked to random-endpoint, reachability, action-mixture, partial-observation, and transport tests. The displayed propositions remain conditional on named assumptions; optional routes receive route-level weight through their protocol-fixed bridges; and finite sampling reaches [SET-ALL] through an independent quantifier bridge.

2. Formal object and primary finite-measure claim

Fix a separately routed maximal qualifying stream i and its measurable complete strict-pre-endpoint truth domain (Ω_i, Σ_i) , and a prospectively declared signed phenomenal representation. On the primary branch, $v_i: \Sigma_i \rightarrow \mathbb{R}$ is a finite signed measure with Jordan decomposition $v_i = v_i^+ - v_i^-$ into mutually singular positive measures. V_i^+ and V_i^- are their finite Jordan masses; their difference is the terminal signed residual.

Primary finite signed-measure branch

$$v_i = v_i^+ - v_i^-, \quad v_i^+ \perp v_i^-.$$

$$V_i^\pm := v_i^\pm(\Omega_i), \quad V_i^{\text{gross}} := V_i^+ + V_i^- = |v_i|(\Omega_i).$$

$$Z_i^* := v_i(\Omega_i) = V_i^+ - V_i^- = L_i^{\text{phen}}(T_i^* -).$$

$$\text{PTVC}_i \Leftrightarrow V_i^+ = V_i^- < \infty \Leftrightarrow Z_i^* = 0.$$

The claim is exact, intrastream, and nontransferable: a residual in one stream cannot be canceled by experience in another. It is also compatible with nonzero gross variation. The mixed-sign case $V_i^+ = V_i^- > 0$ is therefore structurally distinct from a degenerate stream with no phenomenal exposure. Vector-valued, bipolar, nonadditive, or generalized terminal-neutrality branches remain possible, but they are not silently treated as the scalar additive claim. The universal quantifier is conditional on admissibility of the protocol-fixed finite representation; a broader all-stream claim requires a separate domain-coverage or branch-spanning result.

A latent ledger requires an exposure model

On an absolutely continuous subbranch, a latent ledger can be written as an exposure-weighted integral. The general display retains a left-limit term, but on the primary whole-stream branch S_i^θ is the truth-domain origin and $L_i^{\text{phen}}(S_i^\theta -) = 0$; later entry uses the delayed-entry decomposition on the half-open observed interval $[a, b)$. Here $g_{\theta,i}$ and q_i^{exp} are the two factors in the prospectively frozen exposure-density convention. On the primary convention, $g_{\theta,i}$ is a dimensionless time-chart, unity, or ownership factor and q_i^{exp} has exposure-per-time units; on the unity-clock alternative, $q_i^{\text{exp}} \equiv 1$ and $g_{\theta,i}$ carries exposure-per-time units. Their product is experienced-exposure density; neither factor may

duplicate ownership or exposure already encoded by μ_i^{phen} , and v_i^{real} is realized neutral-relative phenomenal value per unit exposure.

Exposure-weighted latent ledger

$$L_i^{\text{phen}}(t) = L_i^{\text{phen}}(S_i^\theta) + \int_{S_i^\theta}^t g_{\theta,i}(s) q_i^{\text{exp}}(s) v_i^{\text{real}}(s) ds.$$

Additivity is the declared representation commitment. Once a signed measure is defined, $v_i(\Omega_i) = v_i^+(\Omega_i) - v_i^-(\Omega_i)$ is an identity. PTVC adds the substantive boundary restriction that the independently defined residual equals zero at the complete endpoint. Constructed zero-integral functions establish mathematical coherence and help design controls; the empirical opportunity is to determine whether independently calibrated natural phenomenal histories satisfy the same boundary.

Primary target: a boundary property of a complete, independently defined stream, distinguished from normalization rules, population averages, and narrative judgments.

Bipolar and generalized branches

The net signed measure need not exhaust phenomenal structure. A bipolar branch allows positive and negative components to coexist, so a common overlap component can increase primitive gross magnitude without changing the net ledger. The net measure then identifies a sharp lower bound on gross variation but no finite upper bound unless overlap is independently bounded. A zero net residual therefore never implies a low-experience or low-suffering stream. Generalized Terminal Neutrality (GTN) is retained for nonadditive or functional-zero representations, but GTN is not relabeled as PTVC unless a valid bridge connects its zero to the additive or bipolar terminal object.

3. Quantifiers, ownership, routing, and falsification

Stream-level adjudication of the universal claim

The primary strict-universal formulation is [SET-ALL]: every separately routed maximal qualifying stream in the protocol-fixed truth-domain class satisfies exact closure. A single gate-valid stream whose nonempty truth-containing residual set—an assumption-guaranteed outer set containing the latent residual—excludes zero is a deterministic counterexample to that protocol-fixed universal branch. A frequentist confidence set supports only its declared coverage statement; a posterior probability is conditional on its protocol-fixed prior and model; and a point estimator remains subject to its uncertainty analysis. Population means, favorable subgroups, or transfers between streams cannot average away a verified counterexample. Probabilistic branches retain their declared probability and error guarantees.

Bearer and routing rules are causal, not clerical

The unit of analysis is a maximal qualifying conscious stream under a prospectively fixed bearer and ownership rule. Splitting one stream after observing its residual, merging streams, dropping weakly observed intervals, or duplicating an event can create apparent balance. Routing must be exhaustive, nonduplicative, time-consistent, and outcome-independent. If individuation is partially identified, report the set over all admissible routings. Unresolved positive and negative contributions enter separate nonnegative ownership-slack accounts and may not cancel each other. Population claims additionally require declared stream-level inclusion probabilities or survey weights; convenient observed streams do not define the target population. The protocol is fixed before quantifying over streams, histories, continuations, or policies.

Modal claims form a partly ordered family

Existence of a completing continuation, positive-probability completion, fixed-policy probability-one completion, almost-sure attainability, limit-sure attainability, topological-support containment, and closure of every admissible continuation are distinct. Almost-sure

attainability requires a single admissible policy attaining probability one; limit-sure attainability permits success probabilities arbitrarily close to one without requiring an attaining policy (Asadi et al., 2025; Chatterjee et al., 2016). Executing a policy additionally requires identifying or constructing it. Every experiment fixes its modality, policy or intervention, continuation class, and evidence object. Realized-policy, all-policy, and all-intervention closure impose different scopes; the latter two require a structural model and an admissible counterfactual domain. No fixed-policy result establishes those stronger scopes or [SET-ALL] without a separate bridge.

Target and route are separate axes

Target support concerns the terminal residual under a validated bridge. Route support concerns an optional mechanism proposed to generate or constrain closure. Their separation strengthens inference: route failure leaves the target open, while route fit advances the target verdict only through a proved entailment.

Evidence remains branch-matched: exact-zero, finite-band, probability, and mechanism claims each use their corresponding support, rejection, and abstention rules.

Decision status remains typed

Public reporting uses a four-axis result vector—TargetStatus, RouteStatus, ComparatorStatus, and InferenceStatus—together with a supporting record of prerequisites, target specification, recovery, and attainment. Table 2 fixes the controlled vocabularies used in this article, and each axis remains independently typed. No composite score may override an adverse target result, rescue a failed prerequisite, conflate target geometries or modalities, or turn route superiority into proof of the terminal law.

4. Representation, neutral origin, and delayed entry

Neutral origin must be calibrated independently

A scalar ledger requires a sign convention, cross-sign comparability, an exposure unit, and a neutral origin. These are measurement hypotheses. Under the admissible affine recoding $v' = a_i + b_i v$ with $b_i > 0$, a stream-constant positive scale preserves a zero residual, whereas on positive-exposure records an additive shift preserves zero only when $a_i = 0$. The origin must therefore be calibrated on nonterminal data or bounded as an uncertainty set before the terminal residual is inspected. The admissible origin class—basis, complexity, envelope, and calibration sample—must be frozen. If it can select a record-specific function that forces zero, the origin is unidentified and the confirmatory result is No adequate estimate (NAE).

Origin sensitivity under affine transformation

$$\tilde{Z}_i^* = a_i E_i^{\text{tot}} + b_i Z_i^*, \quad b_i > 0, \quad E_i^{\text{tot}} = \mu_i^{\text{phen}}(\Omega_i) < \infty.$$

If total exposure is positive, choosing $a_i = -b_i Z_i^* / E_i^{\text{tot}}$ forces the transformed residual to zero. That operation is precisely why completed-record mean-centering is prohibited as confirmatory evidence. A valid cross-sign program perturbs independently validated positive and negative channels, estimates their exchange map, tests state dependence and hysteresis, and freezes the admissible origin set before endpoint fitting.

Study entry and stream origin are distinct

Because most studies start late and stop before the true endpoint, whole-stream claims require one joint truth-containing prefix–observed–suffix object on the same latent scale. It must be narrow enough to decide the protocol-fixed exact or finite-band target; separate component bounds become decisive only through that joint object.

Delayed-entry decomposition

$$\begin{aligned} Z_{i,\text{pre}}(a) &:= L_i^{\text{phen}}(a-), \\ Z_{i,\text{obs}}([a,b]) &:= L_i^{\text{phen}}(b-) - L_i^{\text{phen}}(a-), \\ Z_{i,\text{suf}}(b) &:= L_i^{\text{phen}}(T_i^* -) - L_i^{\text{phen}}(b-), \\ Z_i^* &= Z_{i,\text{pre}}(a) + Z_{i,\text{obs}}([a,b]) + Z_{i,\text{suf}}(b). \end{aligned}$$

The three terms must be constructed jointly so their dependence and simultaneous guarantee are preserved. If either unobserved end is

unbounded or no valid joint truth-containing set can be constructed, the complete-stream result is No adequate estimate (NAE). If a valid joint outer set intersects both the protocol-fixed acceptance set and its complement, the result is indeterminate. A measured branch applies the same accounting to its declared scale without silently promoting it to the latent phenomenal ledger.

Identification gate: a stable sign, origin, unit, and cross-sign scalarization promotes the scalar branch; if one component remains unresolved, the vector-valued alternatives stay available for corresponding tests.

Factorization, scalarity, and gauge freedom

On a density branch, phenomenal contribution can be factorized into exposure and neutral-relative value per unit exposure. Positive predictable rescaling of one factor with inverse rescaling of the other leaves the signed measure unchanged, so exposure and intensity are not separately identified by their product. Clock time, subjective duration, attention, consciousness density, and unity weights must be compared under independent manipulations rather than selected for terminal fit. Raw ratings, ranks, embeddings, factor scores, and logits are not automatically cardinal. Additive-conjoint independence or separability, cancellation, solvability, and held-out consistency test whether one cross-sign scalar is defensible (Luce & Tukey, 1964); bipolar co-occurrence tests whether positive and negative components must instead coexist (Cacioppo & Berntson, 1994; Larsen et al., 2001). An adequate reproducible violation fails the scalar prerequisite; nonidentification or invalidity yields NAE. Prespecified vector, ordered, set-valued, and GTN branches remain open. That representation result is not adverse TargetStatus evidence for every branch.

5. Observation, exactness, and the inference ladder

Let P_i^Z denote the latent terminal-residual law on the declared truth space. Observed reports, physiology, behavior, neural signals, and clinical records arise through a Markov kernel $K_{i,\eta}^{\text{obs}}$ from that latent space to a specified observation space; η indexes instrument, resolution, care context, communication, sedation, cadence, retention, and other declared conditions. The observed law $P_{i,\eta}^{\text{obs}}$ is the kernel push-forward of P_i^Z . Observed variance contraction therefore need not be latent contraction. For longitudinal observations, this kernel also incorporates the declared conditional law of latent histories given the terminal residual. That history law must be specified and validated, or the latent state enlarged; a terminal-residual distribution alone does not determine trajectory observations.

Observation-kernel push-forward

$$P_{i,\eta}^{\text{obs}}(A) = \int K_{i,\eta}^{\text{obs}}(A|z) P_i^Z(dz).$$

Target geometry and exactness levels

- Latent exact target: for a nonempty truth-containing residual set R_i , exact-compatible means $R_i = \{0\}$; exact-adverse means zero is not an element of R_i ; every other relation is exact-indeterminate. A confidence set containing zero is not an exact truth-containing singleton.
- Finite-band target: for a protocol-fixed scientific acceptance set, band-equivalent means the truth-containing residual set is wholly inside it; band-adverse means the two sets are disjoint; partial overlap is band-indeterminate. The familiar $[-\varepsilon, \varepsilon]$ band is one scalar special case. A law-level band-probability claim must separately fix its stream or population law, minimum mass threshold, and error rule in the protocol before outcome inspection.
- Exact atom: the latent terminal law assigns positive mass to zero. Shrinking observed bands do not establish an atom without a validated inverse bridge, deconvolution, or partial-identification argument.
- Law-level quantifiers: probability-one closure under a declared random-stream law and set-universal closure over every qualifying stream are distinct. Neither follows from a large estimated atom mass or a finite sample of compatible streams.

Inference must preserve the object type

Scientific acceptance geometry and epistemic uncertainty are different objects. The acceptance set is fixed prospectively; uncertainty widens the identified region, truth-containing outer set, confidence set, credible set, or sensitivity region rather than that acceptance set (Manski, 2003). A frequentist confidence set is a random set whose protocol-fixed procedure controls repeated-sampling coverage; it is not a deterministic truth-containing set. A Bayesian decision remains prior- and model-conditional, while a deterministic falsifier requires a truth-containing set. Origin, endpoint, routing, missingness, kernel, and numerical uncertainty must be propagated. No geometric relation silently converts finite-band evidence into exact or universal closure.

Equivalence requires affirmative evidence

Equivalence is supported through prospectively specified margins, family-wise error control, boundary-aware asymptotics, mixture alternatives, and model-recovery performance rather than through nonsignificance alone (Schuirmann, 1987; Lakens, 2017; Self & Liang, 1987). A nonidentifying observation map keeps inference at the measured scale until a valid latent bridge is supplied.

Exact atoms require stable inverse recovery

An injective observation map on a protocol-fixed law class does not by itself guarantee stable finite-sample recovery of mass at exactly zero. Diffuse laws concentrated arbitrarily near zero can resemble an atom after noise, rounding, censoring, smoothing, or kernel uncertainty. Recovery studies must therefore include true atoms, near-zero diffuse alternatives, locally discrete laws without a zero atom, mixtures, singular components, quantized channels, observation floors, and perturbed kernels. If the pipeline cannot distinguish those generators at the protocol-fixed resolution, the proper conclusion is a finite band, partial identification, or No adequate estimate. Under a locally smoothing observation kernel, atomless latent laws can concentrate arbitrarily near zero and become arbitrarily close in observed total variation to a true zero atom at every fixed sample size; exactness therefore requires a protocol-fixed separation or structural restriction, not merely more observations under the same unresolved kernel.

6. Random endpoints: a sharp structural discriminator

Exact closure at an independently timed endpoint with a density positive Lebesgue-almost-everywhere imposes a strong and informative terminal signature. Under the assumptions below, for almost every eligible path under the declared conditional reference law, the pre-endpoint ledger equals zero for Lebesgue-almost every time on the interval.

Independent positive-density endpoint proposition

Let E_i be a prospectively declared positive-probability event on which the stream, frozen routing and truth domain, finite ledger, and pre-killing extension exist on a deterministic interval I ; write $P_i^0 = P(\cdot | E_i)$. Under this same law, let $\tilde{L}_i(t-)$ be jointly measurable, $\mathcal{F}_i^L = \sigma\{\tilde{L}_i(s-): s \in I\}$, and T_i^{exo} independent of $\mathcal{F}_i^L$. Assume T_i^{exo} is I -valued, has a Lebesgue density positive almost everywhere on I , and $P_i^0\{\tilde{L}_i(T_i^{\text{exo}}-) = 0\} = 1$.

Independent positive-density endpoint obstruction

$$0 = P_i^0(\tilde{L}_i(T_i^{\text{exo}}-) \neq 0) = \int_I P_i^0(\tilde{L}_i(t-) \neq 0) f_{T_i^{\text{exo}}}(t) dt.$$

The integrand is nonnegative. Tonelli's theorem and the positive density give the product-measure statement: $P_i^0\{\tilde{L}_i(t-) \neq 0\} = 0$ for Lebesgue-almost every $t \in I$; Fubini's section conclusion equivalently gives $\tilde{L}_i(t-) = 0$ for Lebesgue-almost every t on a P_i^0 -probability-one set of paths. If paths are left-continuous P_i^0 -almost surely, equality holds at every interior time; the left endpoint also requires right continuity or a separate value condition. Thus, on the independent positive-density branch, exact closure requires the zero-almost-everywhere structure identified by the proposition.

What the discriminator reveals

The proposition identifies the structure a nondegenerate implementation must supply. Candidate implementations include state-coupled endpoint timing, first-hitting or absorbing boundaries, endpoint atoms or singular support, phenomenally active endpoints, routing change, and failure of the pre-killing extension. Each defines a testable branch with distinctive predictions.

Empirical use

Unexpected and weakly anticipated endpoints are high-pressure tests. The analysis must measure residual dependence between the path and endpoint after prospectively declared covariates, report endpoint surprise, and compare state-coupled hazards, selection, observation shutdown, and random-endpoint bridge rivals. An unplanned endpoint is not thereby exogenous; the protocol-fixed dependence, surprise, and rival checks provide a graded sensitivity analysis, while the independent-

endpoint proposition applies only when its independence premise is separately justified.

What the proposition establishes and which branches it opens

The branch-specific discriminator prospectively localizes routes. Every nontrivial exact-closure model must show structure beyond independent positive-density timing; the candidates above make that testable. Premises remain fixed before residual inspection. Approximate independence gives graded pressure when endpoints sample nonzero-logged times. Ledger-dependent timing identifies a route for direct evaluation; the terminal test adjudicates closure.

Approximate dependence yields a graded pressure test

For $\varepsilon \geq 0$, define $X_i := (\tilde{L}_i(t-))_{t \in I}$, let C_i be prospectively frozen pre-endpoint covariates, and fix a Borel set $J \subseteq I$ with $0 < \lambda(J) < \infty$, where λ is Lebesgue measure. Under P_i^0 , define $g_{i,\varepsilon}(t, c) := P_i^0(|\tilde{L}_i(t-)| > \varepsilon | C_i = c)$, $p_{i,\varepsilon} := P_i^0(|\tilde{L}_i(T_i^{\text{probe}})| > \varepsilon)$, and $\rho_i := E_{P_i^0}[d_{\text{TV}}(P_{X_i, T_i^{\text{probe}} | C_i}, P_{X_i | C_i} \otimes P_{T_i^{\text{probe}} | C_i})]$, where the conditional kernels are evaluated at C_i and $d_{\text{TV}}(P, Q) := \sup_A |P(A) - Q(A)|$. Assume standard-Borel path and covariate spaces, measurable evaluation, the required regular conditional laws, and $f_{T_i^{\text{probe}} | C_i}(t | C_i) \geq m_J > 0$ for λ -almost every $t \in J$, P_i^0 -almost surely.

Approximate endpoint-dependence occupation bound

$$E_{P_i^0}[\int_J g_{i,\varepsilon}(t, C_i) dt] \leq \frac{p_{i,\varepsilon} + \rho_i}{m_J}, \quad m_J > 0.$$

The total-variation event bound changes the probe off-band event by at most ρ_i ; the density floor then gives the display. Consequently, substantial off-band occupation together with rare probe off-band events requires correspondingly large endpoint–ledger dependence after the declared covariate adjustment. Its scientific role is to quantify sensitivity to dependence; independence and SET-ALL remain separate protocol-fixed analyses.

7. Reachability, endpoint support, and solvency

A residual must be cancelable by an admissible future

For a realized stream-owned prefix h_t and prospective endpoint u , define the admissible continuation class $C_i(t, u | h_t)$ without terminal-outcome leakage. Let Cap_i^+ be the supremum attainable future positive contribution and Cap_i^- the magnitude of the infimum attainable future negative contribution. Any continuation that closes the ledger must contribute the negative of its current value, yielding the necessary directional solvency screen:

Directional reachability necessity

$$-\text{Cap}_i^+(t, u | h_t) \leq L_i^{\text{phen}}(t) \leq \text{Cap}_i^-(t, u | h_t).$$

A residual outside this interval makes exact completion at u impossible. An empty continuation class also makes completion impossible, and no capacity supremum is evaluated. An undefined, invalid, or unrecoverable class or decision object yields unavailable or NAE. Passing the interval screen is necessary, not sufficient: the required increment must belong to the actual attainable set, which may have gaps. Compactness can secure attainment of extrema without filling those gaps; completion needs a separate existence or intermediate-value argument under the declared dynamics (Aubin, 1991).

Endpoint support localizes the required structure

If endpoints can occur over broad support, the residual must remain solvent there. Suppose the endpoint counting process has a predictable, locally integrable intensity on the at-risk set that generates an absolutely continuous compensator of finite mass on the analysis interval; assume no simultaneous ledger–endpoint jump. Intensity weighting cannot create a new exact atom at zero from an occupation law that is atomless there. Exact atomic closure instead requires pre-existing atomic occupation, absorption or stickiness, a first-hitting construction, a predictable endpoint atom, a singular compensator, or a justified limiting mechanism. Atomic, singular-compensator, first-hitting, and limiting branches remain separate.

Endpoint adjudication is versioned evidence

Candidate crossing, Estimated endpoint, Confirmed under protocol, Revised endpoint, or No adequate estimate are distinct statuses. Dwell time, return logic, competing events, reliability, and revision rules are frozen before ledger inspection. Later recovery or reclassification preserves the original analysis version and propagates endpoint uncertainty; the last available observation is never forced to be the truth-level endpoint. Reachability measures future capacity; actual selection, mechanism, and completed closure are evaluated through their corresponding protocol-fixed tests.

Endpoint branches are empirical alternatives

Independent positive-density timing predicts the obstruction above. First hitting or absorption can place endpoint mass on a zero surface with a singular compensator rather than an ordinary hazard. Predictable atoms allow mass at discrete boundaries; singular carriers require carrier-specific occupation tests. Endpoint-inclusive branches require supported phenomenal mass and ownership at the boundary. Biological death remains distinct from the truth endpoint. The series separately proposes a biological-death admissibility condition: biological death may not occur before the relevant stream satisfies its declared closure condition. This condition is additional to primary PTV and does not require death when closure is reached. A stream may permanently end after closure while nonconscious biological processes continue, including under artificial life support. Any death-associated contribution follows the declared strict-pre-endpoint or inclusive convention. Routing changes require prospective junction rules and nonduplication. Each branch needs its own evidence; defining an endpoint as a zero hit makes closure true by construction, whereas predicting coincidence with an independently adjudicated endpoint permits a route test.

Unexpected or weakly anticipated natural endpoints are especially informative because a local controller may lose sufficient actionable horizon while a global or state-coupled route may transport. Ethical use is observational: no study may alter survival or care to create endpoint variation.

8. Optional mechanism routes and path-class constraints

PTVC is deliberately mechanism-neutral, allowing reduced reserve-coupled routes, conditioned bridges, absorbing processes, constrained controllers, and other constructions to compete through their distinctive assumptions and predictions. The core target remains independently evaluable across those routes.

Finite variation restricts exact diffusion bridges

A finite signed-measure ledger has finite variation; a continuous version therefore has zero continuous quadratic variation. Suppose an exact path bridge identifies it with a reserve-coupled semimartingale whose continuous local-martingale coefficient is the product of a positive ledger–reserve conversion constant, the remaining reserve, and the diffusion coefficient, while the remaining terms have finite variation. Writing continuous quadratic variation in the displays below, exact path equality requires both increments to vanish:

Finite-variation obstruction to an exact diffusion bridge

$$\begin{aligned} [L_i^{\text{phen}}]_t^c - [L_i^{\text{phen}}]_{S_t^\theta}^c &= 0. \\ [L_i^{\text{rc}}]_t^c - [L_i^{\text{rc}}]_{S_t^\theta}^c &= c_{\text{LU}}^2 \int_{S_t^\theta}^t U_i(s)^2 \sigma_i(s)^2 ds = 0. \end{aligned}$$

Here the product of route-specific reserve and diffusion coefficient is assumed locally square-integrable in time, almost surely, so the stated stochastic integral is defined. Exact path equality then forces their product to vanish for Lebesgue-almost every time, almost surely; wherever reserve remains positive, the diffusion coefficient must vanish in the same sense. A nondegenerate diffusion may remain an approximate, terminal-only, or consequence bridge, but not an exact continuous finite-variation path identity without resolving this mismatch (Protter, 2005). Jumps require separate quadratic-variation accounting.

Reserve and endpoint require a tested bridge

Mechanism-level reserve inference begins with a proxy distinct from ordinary time-to-endpoint prediction. A credible proxy is constructed without terminal-neutrality labels, adds predictive value beyond prognosis, frailty, disease severity, treatment, care setting, device dropout, and observation availability, and transports across sites and perturbations. A separately tested coincidence bridge can then connect route-state exhaustion with the truth-level stream endpoint.

Mechanism-localization discipline

An intervention may change assignment, information, policy, action, physical transition, outcome, retention, endpoint, routing, or observation. The earliest distinguishable change in a conditional law is a candidate causal locus only under the declared identification, timing, measurement-lag, and observation assumptions. In continuous time, causal weighting additionally requires positivity and an appropriate local-independence structure (Røysland, 2011). Downstream outcomes of one alteration must be analyzed with their dependence retained, rather than counted as independent confirmations.

Action-mediated route

Under the protocol-fixed action-first factorization, let h be the predecision history, g the declared continuation regime, and $\mathcal{A}^{\text{do}}(h)$ the measurable admissible action domain. Let π be the fixed policy and $p^{\text{comp}}(a \mid h, g)$ the measurable probability that the complete frozen continuation closes after action a under that regime. F_{comp} denotes the induced joint action-suffix completion event. Assume the policy assigns probability one to $\mathcal{A}^{\text{do}}(h)$. Fixed-policy probability-one completion then holds exactly when p^{comp} equals one for policy-almost every selected action. This is a policy-mass statement; topological-support containment, motivation, and endpoint support require separate assumptions.

$$p^{\text{comp}}(a \mid h, g) := P_g(F_{\text{comp}} \mid h, a)$$

$$\begin{aligned}
\pi(A^{\text{do}}(h)|h,g) &= 1, \\
P^{\pi,g}(F_{\text{comp}}|h) &= \int_{A^{\text{do}}(h)} p^{\text{comp}}(a|h,g) \pi(da|h,g), \\
P^{\pi,g}(F_{\text{comp}}|h) &= 1 \\
\Leftrightarrow p^{\text{comp}}(a|h,g) &= 1 \quad \pi(\cdot|h,g)\text{-almost everywhere.}
\end{aligned}$$

Exact-differential route

A separate route treats the ledger as a path integral of a one-form on a sufficiently rich state space. If the one-form is the differential of a potential Φ on the relevant domain, an open path from S to T has an endpoint-dependent integral, whereas every admissible closed loop has zero integral. Hysteresis, nonzero circulation, or path-dependent residue under frozen coordinates rejects that exact route unless the declared state is enlarged and independently revalidated.

Exact-differential route and loop diagnostic

$$\begin{aligned}
\omega = d\Phi \quad \Rightarrow \quad \int_{\gamma_{S \rightarrow T}} \omega &= \Phi(x_T) - \Phi(x_S). \\
\oint_{\Gamma} \omega &= 0.
\end{aligned}$$

Empirical weight comes from held-out route predictions beyond generic mathematical existence, including the generic ability to condition an ordinary process on completion; the attainability distinctions in Section 3 remain in force.

9. Completion potential and the transport fingerprint

A discriminating near-term route is a prospectively frozen completion potential transported across independent modules.

Completion potential

Let $H_{i,t}$ be the frozen pre-endpoint filtration completed under P_i^0 , with $\mathcal{E}_{i,t}$ and the current-risk indicator $\mathbf{1}_{\{T_i^* > t\}}$ measurable with respect to $H_{i,t}$. For positive-probability $A_{i,\varepsilon}$, require on $\{T_i^* > t\}$ that $P_i^0(A_{i,\varepsilon} | H_{i,t}, T_i^* > t) = P_i^0(A_{i,\varepsilon} | \mathcal{E}_{i,t}, T_i^* > t) = h_{i,\varepsilon}(t, \mathcal{E}_{i,t})$ almost surely; otherwise augment the state or retain history. Nested bands require potentials monotone in ε . The shrinking-band limit is an exact-atom statement only under the stated regular-conditional construction.

Positive-probability completion potential

$$A_{i,\varepsilon} := \{ |Z_i^*| \leq \varepsilon \}.$$

$$h_{i,\varepsilon}(t, \xi) := P_i^0(A_{i,\varepsilon} | \mathcal{E}_{i,t} = \xi, T_i^* > t).$$

$$\lim_{\varepsilon \downarrow 0} h_{i,\varepsilon}(t, \xi) = P_i^0(Z_i^* = 0 | \mathcal{E}_{i,t} = \xi, T_i^* > t).$$

Let E_i^0 denote expectation under P_i^0 . Require $P_i^0(A_{i,\varepsilon} | H_{i,0}) > 0$ almost surely on the declared risk set. The resulting process is a generic nonnegative normalized conditional-probability martingale and supplies calibration checks. PTVC-specific route evidence arises when protocol-fixed held-out relations transport beyond this generic behavior. Its declared domain requires the stated positive-probability condition; probability-zero conditioning uses a declared regular-conditional bridge, density limit, or other justified construction (Doob, 1957; Chetrite & Touchette, 2015).

Normalized completion-event martingale

$$M_{i,\varepsilon}(t) := \frac{P_i^0(A_{i,\varepsilon} | H_{i,t})}{P_i^0(A_{i,\varepsilon} | H_{i,0})}.$$

$$E_i^0(M_{i,\varepsilon}(t) | H_{i,s}) = M_{i,\varepsilon}(s), \quad 0 \leq s \leq t.$$

The overidentification fingerprint

Estimate one potential on training modules without test-endpoint leakage; freeze its state, normalization, tolerance family, support, and observation pipeline; then predict held-out transition dynamics, action odds, endpoint support, terminal-band behavior, and resolution transport. Observational completion potential under P_i^0 and interventional completion potential under a declared policy are different estimands and may not be exchanged without identification and transport assumptions.

A Doob transform provides a generic conditioned-law benchmark; held-out transport beyond adequate conditioned rivals supplies PTVC-specific route support.

Conjunctive differential signature

Specificity comes from a conjunctive pattern of independently tested signatures. The primary proposed conjunctive route package requires reserve-dispersion behavior; two-time covariance and perturbation response beyond bridge rivals; endpoint-surprise calibration; signed-dose compensation with assigned, delivered, experienced, and phenomenal dose kept distinct; residual incremental value; residual compression sufficiency; transport across transition, action, endpoint, terminal-support, resolution, and observation modules; and equal-budget rival failure on common support. Incremental value and compression remain separate claims.

Random-cut complementarity

At a prospectively randomized cut, finite additivity links the accrued prefix and future remainder to the complete residual. The fixed disjoint half-open truth-domain slices assign any cut-time atom once. The identity is bookkeeping; PTVC adds an independently tested zero restriction. The observed remainder must come from post-cut data or a validated terminal-remainder bridge, never subtraction of the measured prefix from the tested terminal target. Use a two-stage within-stream

design with randomized cuts, independent prefix and remainder calibration, stream-specific intercepts, cluster-aware uncertainty, and the protocol-fixed errors-in-variables model below. In the display, u_{ik} and v_{ik} are the protocol-fixed prefix and remainder observation or bridge errors; their joint law across cuts and channels is prospectively specified. Estimate β freely: $\beta=1$ is the additivity gate, equivalently a protocol-fixed prefix coefficient of -1. On the exact direct-partition branch, $\beta=1$, and $\alpha_i=Z_i^*$ follows only when the relevant combined error location is fixed at zero by design or independent calibration. These are structural restrictions, not observed-data identification results. Interpret or test α_i only after a valid errors-in-variables route identifies or bounds the joint error location, scale or covariance, and admissible bias through repeated measurement, validation data, instruments, structural restrictions, or bounded partial identification. Any additional latent residual belongs to a separately protocol-fixed bridge or model-misspecification branch, not the exact ledger identity. A bounded-episode zero intercept is local unless the cut partitions the complete qualifying stream or a protocol-fixed prefix–observed–suffix bridge covers the omitted domain; a pooled zero is a preliminary result that advances through stream-specific adjudication. Repeated cuts are dependent within stream and do not replace stream-specific adjudication. The slope gate tests consistency with the declared additive partition; it does not by itself validate phenomenal additivity. The independently tested terminal restriction supplies the corresponding PTVC test.

Random-cut additivity and stream-specific residual adjudication

$$\begin{aligned}
 Z_i^* &= L_{ik}^{\text{pre}} + L_{ik}^{\text{post}} . \\
 \hat{L}_{ik}^{\text{pre}} &= L_{ik}^{\text{pre}} + u_{ik}, \\
 \hat{L}_{ik}^{\text{post}} &= \alpha_i - \beta L_{ik}^{\text{pre}} + v_{ik}.
 \end{aligned}$$

Randomized remainder and delayed-treatment localization

Randomized assignment supplies a distinct complete-remainder restriction. Let Z denote assignment, L the pre-assignment signed prefix, and τ a common, prospectively fixed treatment-onset time after assignment. On the primary branch, E contains post-assignment contributions before both τ and the truth endpoint; F contains the remaining strict-pre-endpoint contributions at or after τ . If the stream ends before or at τ , $F = 0$. A separately declared inclusive branch adds the terminal contribution J_T :

$$R = E + F, \quad R_{\text{incl}} = E + F + J_T.$$

Exact closure on the declared branch makes its complete remainder equal to $-L$. If assignment is independent of L , remainder distributions agree across arms; stratified randomization requires within-stratum comparisons or standardization to one baseline distribution. None of these statements requires a particular component or biological marker to differ. A closure-enforcing simulation must check that its selection or reweighting preserves the assignment-independence premise.

A localization prediction requires an all-intervention structural model on a common probability space. For the same pre-assignment history h and declared bearer, routing, representation, time cut, and endpoint architecture, suppose primary closure holds under both arms:

$$L(h) + E^a(h) + F^a(h) = 0,$$

$$L(h) + E^b(h) + F^b(h) = 0.$$

$$E^a(h) - E^b(h) = -[F^a(h) - F^b(h)].$$

An independently justified nonzero restriction on the entire later contribution forces an opposite early difference. Immediate treatment effects alone do not supply that restriction when subsequent experience can compensate. The inclusive branch also includes the arm-specific terminal terms in the subtraction. A directional or distributional biological prediction additionally needs a history distribution and

validated observation or viability bridge; the pathwise identity alone does not identify it.

Assignment before a marker but treatment after it is post-assignment and pre-treatment, not automatically retrocausal. With valid randomization and no pre-treatment assignment channel, ordinary local forward-causal models predict no systematic assignment effect. A different prediction requires a declared global-selection, viability, whole-trajectory, or information mechanism. A marker occurring before a qualifying stream begins concerns the developmental extension rather than a measured phenomenal residual.

For a sharp formal fork, consider one candidate continuation d that ends biological life essentially immediately and another that permits ordinary continuation. Let h now denote the complete predecision history. Under the additional biological-death condition and an independent bridge placing the stream endpoint no later than death, let $C_d(h)$ be the complete decision-to-endpoint contribution for the declared branch. A necessary compatibility condition is:

$$L(h) + C_d(h) = 0.$$

The phenomenally silent case $C_d(h) = 0$ requires $L(h) = 0$; this does not make closure sufficient for death or determine a choice probability. A separate route-selector must state how admissibility affects action and what information it uses. If route-to-contingency mapping is independently randomized and locally concealed, an ordinary local policy cannot condition on that mapping. Supplying it secretly to a simulator does not establish biological access. The immediate-terminal fork is a formal and simulation comparison, not an empirical lethal protocol.

10. Rival tournament under equal information budgets

Every candidate and rival receives the same raw variables, observation kernel, missingness inputs, data split, tuning, representation complexity, computation, and predictive-information budget (Blackwell, 1953). Prospectively chosen proper scores compare predictive laws; the multi-gate rubric remains separate. Relative rank is followed by absolute adequacy, recovery, common support, and open-set checks.

Table 1. Equal-information-budget rival models and their primary differential pressure tests.

Rival model and differential pressure test
Regulation / robust perfect adaptation Can mimic: Adaptation, opponent, homeostatic, allostatic, or robust set-point return Primary differential pressure test: Held-out signed integral and terminal residual beyond local set-point recovery
Computational mood Can mimic: Recent expectations and prediction errors create history dependence Primary differential pressure test: Frozen completion potential adds cross-module predictive structure
Flexible baseline Can mimic: Nonzero subject- or state-specific terminal attractor Primary differential pressure test: Compare zero-restricted and nonzero-attractor baselines under held-out proper scoring, calibration, and absolute-adequacy checks
Conditioned bridge Can mimic: Endpoint conditioning pins paths by construction Primary differential pressure test: Score event probability, normalizer, and unexpected endpoints
Endpoint selection Can mimic: Hazard depends on latent state, frailty, care, or prognosis Primary differential pressure test: Joint hazard–ledger fit and endpoint-surprise transport
Observation floor Can mimic: Resolution loss, clipping, sedation, or channel shutdown Primary differential pressure test: Independent channels and calibrated latent deconvolution

Rival model and differential pressure test
Memory kernel Can mimic: Peak–end and duration-neglect reconstruct retrospective balance Primary differential pressure test: Prospective experience ledger retains signal after memory adjustment
Expected neutrality Can mimic: Population or conditional mean is zero with residual variance Primary differential pressure test: Variance/atom/pathwise tests distinguish mean from closure
Open-set generator Can mimic: Unmodeled dynamics fit none of the candidates Primary differential pressure test: Absolute adequacy and no-forced-choice recovery
Ordinary policy control Can mimic: Reward, effort, habit, threat, and viability shape action Primary differential pressure test: Residual term adds held-out calibration beyond equal-information ordinary policy models

Let Q be a prospectively declared rival law and $\mathcal{G}_{i,0}^{ex}$ its baseline information sigma-field. Expected neutrality under Q is a weaker balance claim. With finite second moment, conditional mean zero plus positive conditional variance on a set of positive Q -probability does not imply an atom at zero or streamwise closure.

Expected neutrality is weaker than pathwise closure

$$E_Q[(Z_i^*)^2] < \infty, \quad E_Q(Z_i^* | \mathcal{G}_{i,0}^{ex}) = 0 \text{ } Q\text{-a.s.}, \\ Q(\text{Var}_Q(Z_i^* | \mathcal{G}_{i,0}^{ex}) > 0) > 0.$$

Model ambiguity is admissible. If two mechanisms induce the same distribution of observables under all allowed interventions, the study cannot identify their locus until an intervention or measurement separates them. Generator and verdict recovery are separate. A pipeline may rank the true generator yet issue the wrong verdict, or abstain while generator identity remains ambiguous. Sealed simulations score generator classification, verdict recovery, model ambiguity, open-set detection, and no-adequate-estimate calibration; none compensates for another.

11. Existing evidence and the scientific opportunity

Established neighboring results

Positive and negative affect can be partly separable and co-occur, supporting signed, bipolar, or vector representations (Cacioppo & Berntson, 1994; Larsen et al., 2001).

Adaptation after life events is well documented and heterogeneous (Diener et al., 2006; Luhmann et al., 2012). Opponent-process, allostatic, incentive-salience, interoceptive-prediction, homeostatic reinforcement-learning, computational-mood, and robust integral-feedback systems generate rebound, history dependence, dissociations among liking, wanting, and learning, and measurable forms of compensation (Solomon & Corbit, 1974; Koob & Le Moal, 2001; Berridge et al., 2009; Barrett & Simmons, 2015; Keramati & Gutkin, 2014; Eldar et al., 2016; Keren et al., 2021; Barkai & Leibler, 1997; Briat et al., 2016; Aoki et al., 2019). These affective findings and biological or computational control results supply distinct precedents and rival models; biomolecular adaptation is not itself evidence of a phenomenal-valence mechanism.

Experienced and remembered utility can diverge through duration neglect and peak–end weighting (Fredrickson & Kahneman, 1993; Kahneman et al., 1997). This divergence makes prospective measurement and explicit memory rivals essential. PTVC extends that foundation into a new integrative question: whether complete-stream dynamics exhibit an exact terminal boundary condition.

Plausibility signals and evidential roles

- Hedonic adaptation, biological set-point control, and terminal change provide mechanism precedents, comparator models, and endpoint-study opportunities whose PTVC-specific weight is determined by protocol-fixed, branch-matched bridges.
- Constructed zero-integral functions, completion potentials, bridges, and Doob transforms establish mathematical coherence and generate testable predictions; held-out transport determines their empirical weight.
- Retrospective life evaluation provides informative evidence through a memory-and-demand observation kernel and can be compared directly with prospective experience ledgers.
- Endpoint proximity and changing channel variance become informative when modeled jointly with frailty, treatment, sedation, device failure, censoring, and observation floors.

How independent convergence builds confidence

The neighboring literatures contribute premises, instruments, comparator mechanisms, datasets, perturbations, and bias structures. A dependency graph makes shared samples, instruments, preprocessing, cultural ancestry, and observation channels visible so independent convergence receives appropriate weight. Confidence grows most when one frozen relation transports through distinct paths with different expected biases: prospective experience and retrospective memory; human and nonhuman tasks; survey, physiology, behavior, and neural measurement; ordinary and terminal horizons; observational cohorts and randomized perturbations. This triangulation strategy gives more weight to methodologically independent evidence than to raw citation count (Lawlor et al., 2016; Munafò & Davey Smith, 2018).

Endpoint, horizon, and consciousness literatures

Perceived future time reorganizes motivation and emotional priorities, making disclosed horizon a strong planning rival to actual endpoint coupling (Carstensen et al., 1999; Carstensen, 2006). Time-to-death studies report heterogeneous changes in affect, symptoms, depressive burden, and psychological or social well-being near death (Vogel et al., 2013; Seow et al., 2011; Petkus et al., 2025; Wang et al., 2025). These findings support endpoint-aligned study design and provide baselines for testing declared preterminal signatures. Complete-stream measurement and terminal-zero adjudication require the additional representation and endpoint bridges developed here.

Behavioral nonresponse can conceal cognition; last report, last response, biological death, and last qualifying experience need not coincide (Casali et al., 2013; Bodien et al., 2024; Parnia et al., 2023). Comparative consciousness research cautions against limiting the eligible class to verbal humans (Birch et al., 2020). These literatures justify multimethod endpoint intervals with explicit uncertainty and provide concrete tools for testing stream individuation and eligibility across species.

Recent multicenter multimodal work in disorders of consciousness supports combining behavioral assessment, neuroimaging, and electrophysiological markers (Manasova et al., 2026). Extending this approach to PTVC requires separately validated valence and endpoint bridges; the published diagnostic and prognostic findings do not themselves identify phenomenal closure.

12. Measurement, missingness, and endpoint validity

Representation before accumulation

A confirmatory instrument must establish sign, origin, scale behavior, temporal resolution, exposure weighting, and measurement invariance across the populations and states being compared (American Educational Research Association et al., 2014; Krantz et al., 1971; Meredith, 1993). Method-diverse channels should be treated as separate fallible measurements of a latent object, not averaged until their error structure is modeled. Multitrait–multimethod validation and negative-control channels test whether common arousal, salience, motor capacity, or demand explains apparent convergence (Campbell & Fiske, 1959).

Observation coverage completes the estimand

Scheduled, randomized, participant-triggered, clinician-triggered, safety-triggered, endpoint-triggered, and device-failure observations retain separate marks. Crisis-enriched or terminally enriched samples alone do not identify a time- or exposure-weighted complete-stream integral.

Recovery requires an outcome-independent coverage component, a justified sampling-process correction with positivity, scientifically valid bounds for unsampled intervals, or an explicit partial-identification verdict.

Missing not at random is a central rival

Communication failure, fatigue, delirium, sedation, symptom severity, withdrawal, care transitions, and device loss can depend on latent valence and endpoint proximity. A 12-month burst-sampling study demonstrates that longitudinal completion and attrition patterns can be modeled explicitly and supplies a useful design template for the present missingness analysis (Prochnow et al., 2025). Selection, pattern-mixture, joint longitudinal-event, inverse-probability, and bounding approaches should be compared across prospectively specified sensitivity ranges rather than relying on complete-case or last-observation-carried-forward defaults (Rubin, 1976; Scharfstein et al., 1999; Little & Rubin, 2019).

No-lookahead and reactivity

Endpoint classifiers, smoothed latent states, reserve proxies, and learned representations may not use future observation availability or terminal labels unless the analysis is explicitly retrospective. The burden and emotional effect of measurement enter the trajectory as intercurrent events. Every cumulative proxy must pass a no-neutrality-by-construction simulation.

Positive observation probability supports recovery only with an identified observation process and measurement bridge. Valid finite bounds for unobserved exposure generally provide partial identification; point identification requires the resulting target set to be a singleton.

Replicate channels and recoverability

Method-diverse channels can identify or bound a latent distribution only under explicit independence, loading, location, moment, and support assumptions. Two-replicate identification in the spirit of Kotlarski (1967) is available only under its own additive-error and independence conditions; analyst separation and instrument diversity do not prove stochastic independence. Every bridge is first stress-tested in sealed simulations containing assumption violations, shared causes, response shifts, shutdown, and state-dependent reporting. A favorable observed result first constitutes measured-scale evidence and advances to latent PTVC evidence as the protocol-fixed latent gates are satisfied. The Day Reconstruction Method improves structured within-day coverage but remains a retrospective reconstruction whose recall and selection process belongs in the observation model (Stone et al., 2006).

Randomized ecological momentary assessment (EMA) studies show that prompt schedule and response format can alter measured affect distributions or dynamics. Prompt cadence, response format, anchors, and interface defaults therefore belong inside the observation kernel and require frozen or calibrated transport rules (Langford et al., 2026; Haslbeck et al., 2025).

13. Confirmatory decision structure and model recovery

Fix deterministic verdict rules before outcome access. Table 2 is the required confirmatory reporting record: it separates the four-axis public result vector from supporting prerequisite, specification, recovery, and attainment fields, and its conjunctive gates prevent one passed module from masking another’s failure.

Table 2. Required confirmatory reporting record and four-axis public result vector.

Field	Protocol-fixed vocabulary and adjudication rules
Prerequisite Status	Valid / failed / unavailable — Representation and sign; neutral origin; protocol-fixed scalar or non-scalar branch; exposure rule and clock; qualifying bearer; ownership and routing; complete strict-pre-endpoint domain; truth-level endpoint; observation and missingness; measured-to-latent bridge. A failed or unavailable prerequisite yields No adequate estimate / invalid gate and blocks downstream promotion.
Target Specification / TargetStatus	Type: exact residual / finite band / atom at zero. Modality: [SET-ALL] / [P1] (probability-one) / [POLICY-ALL] / [SUPPORT] / [CONTINUATION-ALL] / [FINITE-CLASS]. TargetStatus uses the controlled verdict lexicon: Equivalent; Different / off-band; Informative but inconclusive; No adequate estimate / invalid gate; or Target-adverse, matched to the specified target. Exact-atom evidence remains nested and branch-matched.
Recovery Status	Record generator recovery, scientific-verdict recovery, and absolute adequacy. Recovery or availability failure yields No adequate estimate / invalid gate; failure of the declared open-set check yields Open-set inadequate.

Field	Protocol-fixed vocabulary and adjudication rules
RouteStatus / Comparator Status	RouteStatus and ComparatorStatus use the controlled lexicon: Route-favored; Rival-favored / Route-failed; Model-ambiguous; Open-set inadequate; or No adequate estimate / invalid gate. Target-adverse remains a TargetStatus result. Equal-budget scoring, calibration, absolute adequacy, common support, and open-set checks govern promotion.
Attainment Status	fixed-policy probability-one / almost-sure attainable / limit-sure only / positive-probability only / unavailable - These route-policy quantifiers do not replace TargetSpecification modality.
Inference Status	identified-set / frequentist / Bayesian / sensitivity-based — Preserve the evidence-object type, assumptions, coverage or prior basis, and sensitivity guarantees. Controlled public verdict labels remain on their corresponding axes, with no cross-promotion.

Recovery before interpretation

Synthetic and semi-synthetic studies span PTVC routes, expected-neutrality laws, diffuse and atomic residuals, flexible baselines, conditioned bridges, state-dependent hazards, regulatory and computational models, memory reconstruction, selection, censoring, heavy tails, stochastic volatility, and observation floors.

A correctly computed empty inversion or feasibility set is not NAE and receives its protocol-fixed object-matched state—model-incompatible, bridge-failed, route-failed, or another named rejection. NAE is reserved for unavailable, undefined, invalid, or uncomputed objects. A protocol-fixed recovery matrix reports generator classification and scientific verdict by generator class, including worst-class performance. The pipeline controls false favorable support and false counterexample declarations.

Negative controls and open-set alternatives

Negative controls test leakage and common-cause explanations (Lipsitch et al., 2010). If every offered model fails absolute adequacy, report Open-set inadequate; adequate but inseparable alternatives yield Model-ambiguous. Unavailable or invalidly recoverable target information yields No adequate estimate / invalid gate. Open-set generators prevent forced selection among misspecified candidates. Proper scores and calibration accompany scientific margins rather than override these distinctions (Gneiting & Raftery, 2007).

Protocol versioning preserves every result: later refinements remain traceable to the version under which each gate and counterexample was evaluated.

Unresolved cases remain visible

Unresolved bearer, endpoint, observation, or bridge cases cannot disappear from the denominator. Use one fixed target, modality, eligible frame, and weighting rule. The displayed interval is a sharp deterministic bookkeeping bound only when favorable and adverse classifications are truth-containing under the same gate system. Confidence-set versions require a simultaneous coverage statement; posterior and sensitivity versions retain their protocol-specified interpretations. Without such a guarantee, report classification bookkeeping rather than a bound on the true violation prevalence.

Transparent unresolved-case bound

$$W_E := W_F + W_A + W_U > 0, \quad \frac{W_A}{W_E} \leq p_{\text{viol}} \leq \frac{W_A + W_U}{W_E}.$$

Let W_F be weighted favorable classified mass, W_A weighted adverse classified mass, W_U weighted unresolved or gate-invalid mass, W_E total eligible mass, and p_{viol} the eligible-population violation fraction. This interval is deterministic bookkeeping, not a confidence or posterior interval unless the protocol-fixed simultaneous inferential guarantee supplies that interpretation.

14. Safe nonterminal program: calibration before terminal claims

The first experiments build the evidential foundation through safe nonterminal studies of representation and local route predictions. Successful modules prepare the program for later complete-stream PTVC testing while initially using nonterminal populations and reversible interventions.

Stage 0 — protocol fixation, public preregistration, and rigorous design

Freeze the target, measurement, route, rival, scoring, error-control, abstention, retirement, and software choices listed in the protocol. Conceal generator labels from analysts and adjudicators until the simulation pipeline is locked. Where roles permit, conceal empirical outcomes until the empirical pipeline is locked. Document the two procedures separately.

Stage 1 — representation and local complementarity

Use safe bounded episodes with randomized benign assignments across valence modalities, controllability, predictability, duration, and horizon, drawing on micro-randomized trial design (Klasnja et al., 2015). Under consistency, well-defined versions, positivity, and the declared interference structure, randomization identifies assignment effects—not delivered, experienced, or latent phenomenal dose. Those dose variables remain separately calibrated with explicit noncompliance and measurement bridges. Test whether an independently measured future response area complements accrued area at randomized cuts against equal-budget ordinary rivals. A transported local cancellation result would support episode-scale route calibration and prepare the next prospectively specified terminal test.

Stage 2 — frozen completion-geometry transport

Estimate one completion potential on training modules and freeze it. On held-out modules, test transition calibration, conditional increments, action odds under randomized information, future-area prediction, residual compression sufficiency, and resolution transport.

Off-support actions return partial identification or No adequate estimate. Model selection and representation learning remain inside the training loop.

Promotion criterion

Promotion requires generator and verdict recovery, stable origin and measurement, held-out calibration, superiority to equal-budget rivals, correct negative-control behavior, and replication across a site or channel. Proper scores evaluate predictive distributions; the conjunctive vector remains a decision rubric. Passing supports only the local route and progression to terminal studies.

Immediate analyses in existing data

Before new endpoint studies, existing longitudinal resources can immediately test foundational pipeline components. A measured ledger proxy must be built without future endpoint information. At each prospectively specified landmark, all representations, covariates, competing-risk forecasts, and tuning are fit out of fold using data available by that landmark; realized time and cause are then scored before route-by-surprise tests. Held-out compensator-residual and time-rescaling diagnostics assess calibration under a construction matched to the protocol-specified ordinary, marked, or cause-specific, terminating, and right-censored endpoint process (Cox, 1972; Andersen et al., 1993; Brown et al., 2002). Cause-specific intensity, cumulative incidence or event probability, and interventional endpoint risk are distinct estimands. Thus the test is whether one frozen relation outpredicts selection, local control, prognosis, and hidden-state rivals—not whether hazard merely declines near zero.

Terminal-dispersion analyses must decompose observed variance into latent state, observation error, and their covariance; narrowing can reflect true contraction, censoring, scale compression, sedation, or channel loss. Cohorts with dense affect and later global evaluation can separate experienced value from peak-end, duration-neglect, and narrative-memory rivals. Cross-dataset analyses should retain

dataset-specific measurement models and use leave-one-dataset or leave-one-country-out transport rather than force a common coefficient. Causal transport requires explicit invariance assumptions rather than coefficient similarity alone (Pearl & Bareinboim, 2014).

Analyses comparing anticipated with unexpected endpoints freeze a competing-risk forecast at a prospectively specified landmark, score realized time and cause, and only then test route-by-surprise interactions.

Three high-value experiments

First, prospectively specified nonanticipating stopping kernels randomized from frozen history probe occupation functionals on a protocol-fixed basis. Full recovery of unrestricted basis coefficients requires separation; particular functionals or informative bounds can remain available without it. Require well-defined intervention versions, consistency, no interference or a declared interference map, known assignment or conditional exchangeability of future potential outcomes, positivity on common support, and declared observation and censoring models. Report basis, rank, condition number, and null space; distinguish physical stopping from disclosed horizon. Ill-conditioned recovery requires stability checks; report identified functionals or valid bounds, and use NAE when neither is available for the declared target. Second, random-cut complementarity with benign appetitive and aversive micro-interventions separates bookkeeping from causal compensation while assigned, delivered, experienced, and phenomenal dose remain distinct. Third, freeze one completion potential across transition, action, endpoint, and terminal-support modules. All three use sealed simulations, open-set generators, and local claim ceilings.

Comparative route and future-regime tests

A *Drosophila* test should begin only after a declared route model predicts a directional or distributional history-by-choice effect. After adult eclosion, independently randomized experimental units receive welfare-compatible resource or opportunity histories under balanced

developmental and genetic conditions. These histories are operational manipulations, not measurements of positive or negative phenomenal totals. Following a fixed common-condition interval, measure the first choice before feedback between routes leading to model-relevant continuing environments. Randomize and counterbalance the route-to-contingency mapping while matching immediate cues and motor costs. Code the primary contrast by assigned downstream contingency, with physical side separate and a control-derived baseline rather than an assumed 50/50 split; retain nonchoice and observation loss as distinct outcomes.

Hunger, locomotion, age, sex, mating status, circadian time, associative history, differential survival, batch, and cues enter the prespecified design or rival pathways, not an automatic covariate-adjustment list. Satiation-dependent foraging, learned resource preference, and state-dependent ambiguous-cue responses supply ordinary explanations (Landayan et al., 2018; Zrelec et al., 2013; Deakin et al., 2018). Vial- or cohort-level assignment makes that unit the experimental replicate, with flies nested within it; sample-size justification, masking, exclusions, and interference handling follow the actual design (Lazic et al., 2018; Percie du Sert et al., 2020). Cue checks report their detection sensitivity. A continuing-environment analogue needs its own derived prediction: removing the terminal contingency does not automatically preserve the formal discriminator. Behavior alone does not establish a qualifying phenomenal stream.

A delayed-treatment extension may assign eggs or independent egg batches to future adult-condition schedules under matched pre-treatment physical conditions. Freeze one primary post-assignment/pre-treatment marker and prediction; for eclosion, adult treatment starts only after its observation window closes. The confirmatory analysis follows assignment. A model depending instead on the regime actually realized must specify a causal or partial-identification strategy for cancellation, crossover, non-delivery, and failure to reach adulthood, rather than condition naively on later treatment receipt (Hernán & Robins, 2020).

Core PTVC does not itself require a hatching, survival, eclosion, or choice difference.

The sealed benchmark should include null finite-sample imbalance, flexible later compensation, early stream endings, endpoint-inclusive variants, cue or allocation leakage, parental and batch effects, selection, nonchoice, learning, and regime changes. It should assess generator and verdict recovery, absolute adequacy, and false-positive calibration. Permutation and post-outcome placebo labels follow the actual allocation ratios and blocks; placebo labels are generated independently of outcomes, do not assign treatment, and check analysis calibration rather than exclude original physical leakage. A biological prediction advances only with its independently justified localization or viability bridge and scientific and welfare review. The formal terminal fork remains computational; empirical conditions remain welfare-compatible.

15. Endpoint-aligned extension and complete-stream adjudication

Observational bridge studies establish the path to terminal tests

Existing longitudinal cohorts and low-burden clinical data can evaluate measurement invariance, time-to-endpoint alignment, missingness models, endpoint revision, residual variance, and rival recovery. These studies establish feasibility, informative bounds, and recoverability, and can test whether declared signatures remain distinguishable from symptoms, function, prognosis, and channel shutdown. Identification of phenomenal closure additionally requires the complete-domain and latent-measurement bridges.

Prospective endpoint architecture

A terminal protocol needs serial multimethod assessment, an independent endpoint committee, immutable status history, competing-event and return logic, device and communication-quality records, medication and care context, and a joint endpoint–observation–missingness model. The endpoint predicate is fixed without ledger outcomes. Observation burden is minimized and never allowed to interfere with treatment or comfort.

Primary estimands

- The latent terminal residual compatibility set, including origin, endpoint, prefix, suffix, observation kernel, measured-to-latent bridge, and numerical uncertainty.
- Terminal residual atom location and mass versus no-atom-at-zero alternatives, including off-zero atoms, lattice or quantized, singular-continuous, absolutely continuous, and mixed laws.
- Endpoint-dependence and hazard-weighted occupation diagnostics, including unexpected-endpoint stress tests.
- Transport of the frozen completion potential and local signed-dose calibration into the endpoint-aligned module.

Evidence requirements for complete-stream adjudication

Complete-stream adjudication applies the joint prefix–observed–suffix rule from Section 4: the truth-containing object, plus an adjudicated endpoint or validated terminal-remainder bridge, must locate the full latent residual against the protocol-fixed exact or finite-band target. Until then, separate bounds, protocol endpoints, and measured-scale results remain staged inputs.

Clinical ethics govern scientific ambition: studies preserve care and comfort, and no intervention may induce harm, delay care, manipulate dying, or treat PTV C as guidance before empirical adjudication.

Endpoint surprise and promotion

A prospective program distinguishes disclosed horizon, predicted endpoint hazard, and realized time and cause. One route may work only when participants can plan; another may predict structure under concealed or unexpectedly rapid endpoints. Promotion requires the same measurement, recovery, comparator, common-support, and truth-bridge gates used to declare adversity; favorable and adverse claims receive symmetric scrutiny.

16. Exactness, atoms, and symmetric error control

Exact zero is a precise boundary property. Its evaluation separates location, spread, atom mass, and the observation process while accounting for finite resolution, measurement error, smoothing, selection, and model flexibility.

Atom at zero versus no atom at zero statistical branch

$$P_i^Z(dz) = \pi_{0,i} \delta_0(dz) + (1 - \pi_{0,i}) G_i(dz), \quad G_i(\{0\}) = 0, \quad 0 \leq \pi_{0,i} \leq 1.$$

For stream i , the no-atom-at-zero alternative G_i satisfies $G_i(\{0\}) = 0$ and may contain off-zero atoms, lattice or quantized structure, singular-continuous or absolutely continuous mass, and mixtures. The finite-band branch asks whether $P_i^Z([- \varepsilon, \varepsilon])$ meets a protocol-fixed threshold; the atom branch asks $\pi_{0,i} > 0$; probability-one closure asks $\pi_{0,i} = 1$. A location-free rival estimates stream-specific atom location a_i and mass $\pi_{a,i}$; its remainder $G_{i,a}$ assigns zero mass to $\{a_i\}$. Atom existence and zero location are therefore scored separately. [SET-ALL] is the stronger deterministic universal claim.

Boundary-aware inference

Both $\pi_{0,i} = 0$ and $\pi_{0,i} = 1$ are nonregular boundaries, so ordinary interior asymptotics may fail (Self & Liang, 1987); at $\pi_{0,i} = 1$ the remainder law G_i is unidentified. Recovery should include parametric bootstrap or valid branch-specific methods, atomless laws with high density near zero, near-unit mixtures, location-free atoms, singular alternatives, rounding, heteroskedastic error, missingness, and finite-sample stress tests.

Nested-resolution discipline

Prospectively nested bands may increase discrimination only while the latent target, acceptance sets, bridge family, recovery guarantees, and limiting rule remain fixed. Passing a smaller observed band after changing the instrument or model does not constitute convergence to exactness. Independent channels and transport across resolution are more informative than repeated analyses of one kernel.

Symmetric error-control budget

The confirmatory multiplicity budget is symmetric: it controls false favorable support across representation, origin, routing, endpoint, missingness, route, rival, tolerance, and subgroup searches, and false counterexample declaration across streams, endpoints, instruments, and models. Adaptive choices remain exploratory or inside training. A protocol-fixed deterministic falsifier, once verified on held-out data with its truth bridge intact, is not weakened by unrelated multiplicity; a noisy search for such a falsifier remains subject to multiplicity control. Promotion and retirement each require their protocol-fixed outer decision set.

17. Falsification, safeguards, ethics, and research agenda

Results that would narrow or reject a protocol-fixed PTVC claim

Because the program is built to learn, clear counterevidence would be a valuable scientific result. A deterministic, nonempty truth-containing residual set for a gate-valid stream that excludes zero refutes the protocol-fixed exact set-universal class. Other branch-matched results include endpoint independence with demonstrably nonzero ledger occupation; imbalance outside validated remaining capacity; reproducible scalar or bipolar representation failure; origin flexibility that manufactures closure; cross-channel incompatibility; and repeated route failure against adequate equal-budget rivals. Frequentist adverse decisions preserve coverage and multiplicity; posterior adversity remains prior- and model-conditional. A better-fitting no-atom model weakens the exact-atom branch; memory-only closure favors memory rivals; missing-not-at-random or endpoint sensitivity can remove terminal support; and adequate-rival parity removes differential route support.

Known challenges and their research solutions

Phenomenal sign, cardinal or interval scale, neutral origin, exposure, stream individuation, ownership, and truth-level endpoint are demanding latent constructs. Complete histories are rarely observed, endpoint and observation processes are entwined, and some models may remain partially identified. The architecture turns these challenges into explicit measurement, partial-identification, transport, and decision problems, allowing each result to retain a clear representation, bearer class, species, culture, endpoint cause, and resolution.

Ethical boundary

Ambitious research is trustworthy only when care comes first. PTVC is a research hypothesis; clinical, legal, moral, and policy decisions retain their independent evidentiary and ethical foundations. It cannot justify harm, denial of care, discounting suffering, or treating another person's experience as transferable compensation. Human research must follow

applicable requirements, preserve informed consent and withdrawal, protect privacy, minimize burden, and receive independent oversight (World Medical Association, 2024). Animal work follows applicable welfare requirements and replacement, reduction, and refinement principles, with transparent reporting (Percie du Sert et al., 2020). A Law-specific study must not create a harmful terminal condition to sharpen a contrast; the severe terminal fork stays formal and simulated.

What should materially change belief

A strong favorable pattern would combine a frozen valence representation, stable origin and exposure rule, method-diverse channels, one joint truth-containing prefix–observed–suffix object, a validated endpoint and observation model, contraction of the latent compatibility set under linked resolutions, one completion potential transported across modules, superior held-out performance against adequate equal-budget rivals, correct endpoint-surprise behavior, and independent replication. Such a result would constitute strong support for the protocol-fixed class, branch, and resolution while defining the next quantifier-bridge challenge.

These criteria make belief revision constructive: favorable signatures gain force as protocol-fixed bridges pass, while resolved ambiguities narrow the live explanations. Near-zero means, adaptation, terminal decline, cultural recurrence, fitted bridges, and partition slopes gain evidential weight through independent measurement, transport, and equal-budget rival comparison; nonsignificance alone remains nonadjudicative.

Near-term prerequisite and differential predictions

One near-term check is Section 9’s random-cut restriction: $\beta = 1$ (prefix slope -1) under the declared representation and an identifying or bounding error model. Agreement supports consistency with the additive partition, not its full phenomenal validity or PTVC closure. A nonzero intercept supports nonclosure only when its error location and complete-stream domain are identified or validly bounded; for a

bounded episode, the conclusion remains local. A failed slope directs review to representation, sampling, or the observation bridge.

Early publishable milestones

A valuable paper can materially advance PTVC by validating one frozen representation, origin rule, exposure clock, observation kernel, missingness model, endpoint interval, or local route in a protocol-fixed class. It should report generator recovery, control performance, uncertainty, and transport. Its immediate contribution is a robust measurement or local mechanism result and a rigorous foundation for terminal studies.

Cumulative decision bookkeeping

Each result enters a versioned evidence ledger recording the target and modality, bearer and routing, representation, origin and exposure, endpoint, observation and missingness, reference law and support, tolerance or atom claim, rivals and budget, estimand and proper score, recovery, decision, and claim ceiling. Outcomes downstream of one perturbation remain one evidence family. Promotion requires its held-out gate; no index offsets a failed target or truth bridge. Replication changes a principal bias—site, instrument family, analyst team, endpoint process, species, or intervention—while transporting the same frozen object.

Promotion, retirement, and open challenge

The project should maintain a public generator bank that includes supportive, rival, mixed, singular, near-zero, measurement-failure, missing-not-at-random, and none-of-the-above worlds. New routes enter only with independent equations and prospective retirement rules. A route retires after reproducible recovery failure, failure of its defining equation under adequate power, or loss to an equally resourced rival across its declared support. A target branch retires when a gate-valid counterexample satisfies its protocol-fixed falsifier. Escalation to vulnerable or endpoint-proximal populations occurs only after safer

modules pass, the additional information cannot be obtained otherwise, independent oversight approves the burden, and the protocol cannot influence survival or care. The same infrastructure must make adverse, ambiguous, and no-adequate-estimate outcomes as publishable as favorable ones.

Ordered research agenda

The agenda begins with independent formal review and generator-recovery challenges, followed by representation, observation, and bounded route studies. Random-cut, signed-dose, horizon, and completion-potential work can proceed alongside comparative-route and future-regime simulations when their own prerequisites hold. The latter designs advance to welfare-compatible animal studies only after their model and design gates pass; animal testing is not a prerequisite for unrelated human or theoretical branches. Endpoint-aligned cohort analyses, low-burden multimethod studies, and direct complete-stream tests follow the measurement and endpoint evidence each requires. Each stage records outcomes, versioned protocols, reproducibility materials, and predetermined promotion and retirement rules.

Program-level conclusion. PTVC converts a broad conjecture into a typed boundary hypothesis that can be measured, compared, supported, narrowed, or refuted. It brings separable positive and negative affect, opponent and adaptive dynamics, integral feedback, history-sensitive mood, horizon-dependent regulation, terminal trajectories, consciousness assessment, and latent-state and endpoint inference into one testable frame. The framework specifies how future studies could determine whether a declared signature favors a terminal-boundary model, a rival, or neither. A gate-valid counterexample, a transported signature, or a principled No adequate estimate would each advance the science and invite interdisciplinary collaboration, replication, and appropriately staged support.

18. Evaluation and reproducibility standard

Evaluation asks whether the formal target and research architecture meet internal-validity, claim-ceiling, comparative-testability, ethical-feasibility, and reproducibility standards. Meeting them makes PTVC scientifically assessable and creates a disciplined path for empirical support to accumulate through prospectively specified studies.

Modular, noncompensatory evaluation

Evaluation keeps five modules separate: formal coherence; measurement and identification; endpoint and ownership validity; recovery and error control; and ethics and reporting. Strengths and limitations therefore remain local to the affected branch, assumption, estimand, or verdict. A limitation narrows or retires that branch; a blocked complete-stream object receives NAE until its bridge is supplied.

Minimum reproducibility capsule

Every empirical study derived from this framework should archive a timestamped protocol; claim, formula, bearer, routing, endpoint, observation, missingness, generator, data-split, software, seed, and code records. Data provenance, exclusions, transformations, linkage, access restrictions, and time windows must be documented. Outputs should include machine-readable decisions, recovery matrices, acceptance geometry, planned sensitivities, and public-artifact hashes. Publish executable synthetic fixtures, schemas, code, and decision records where lawful and safe; use governed access or documented substitutes for restricted materials and state which independent checks remain possible.

Independent audit and discrepancy handling

Before outcome access, an independent audit checks object typing, formula order, endpoint and cut randomization, no-lookahead, action domains, policy mass, rank, weighting, multiplicity, and open-set behavior. Corrections are versioned without replacing the original protocol, output, or verdict. Distributed formats and machine-readable

sources must match the displayed mathematics and claims. Production fixes leave the scientific model unchanged; scientific changes receive a new version.

Promotion, retirement, and publication outcomes

Publication outcomes follow Sections 14 and 17's protocol-fixed promotion and retirement rules. Every protocol-defined outcome remains publishable; verified counterexamples remain attached to the tested version, finite-resolution compatibility remains resolution-limited, and no target-defining element may move after inspection.

Declarations

Author: Stephen Takowsky

Funding. No external funding.

Copyright. Text and compilation copyright © 2026 Law of Fairness LLC. All rights reserved.

Competing interests. Stephen Takowsky originated PTVC, founded the Law of Fairness Project, and owns or controls rights in related publications and project materials.

Data, code, and materials. The operative definitions, propositions, and decision rules used here are stated in this article with their cited mathematical and methodological sources. No new participant dataset is reported. Studies derived from the framework should archive and share code, synthetic fixtures, schemas, and decision records under their prospectively specified ethics, privacy, licensing, and version-control plans.

Ethics statement. This work involved no human or animal participants and used no participant data.

Correspondence. team@lawoffairness.com.

References

- American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). Standards for educational and psychological testing. American Educational Research Association.
- Andersen, P. K., Borgan, O., Gill, R. D., & Keiding, N. (1993). Statistical models based on counting processes. Springer.
<https://doi.org/10.1007/978-1-4612-4348-9>
- Aoki, S. K., Lillacci, G., Gupta, A., Baumschlager, A., Schweingruber, D., & Khammash, M. (2019). A universal biomolecular integral feedback controller for robust perfect adaptation. *Nature*, 570, 533–537.
<https://doi.org/10.1038/s41586-019-1321-1>
- Asadi, A., Chatterjee, K., Saona, R., & Shafiee, A. (2025). Limit-sure reachability for small memory policies in POMDPs is NP-complete. *Proceedings of Machine Learning Research*, 286, 238–256.
<https://proceedings.mlr.press/v286/asadi25b.html>
- Aubin, J.-P. (1991). Viability theory. Birkhäuser.
- Barkai, N., & Leibler, S. (1997). Robustness in simple biochemical networks. *Nature*, 387, 913–917. <https://doi.org/10.1038/43199>
- Barrett, L. F., & Simmons, W. K. (2015). Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7), 419–429.
<https://doi.org/10.1038/nrn3950>
- Berridge, K. C., Robinson, T. E., & Aldridge, J. W. (2009). Dissecting components of reward: “Liking,” “wanting,” and learning. *Current Opinion in Pharmacology*, 9(1), 65–73.
<https://doi.org/10.1016/j.coph.2008.12.014>
- Birch, J., Schnell, A. K., & Clayton, N. S. (2020). Dimensions of animal consciousness. *Trends in Cognitive Sciences*, 24(10), 789–801.
<https://doi.org/10.1016/j.tics.2020.07.007>
- Blackwell, D. (1953). Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2), 265–272.
<https://doi.org/10.1214/aoms/1177729032>
- Bodien, Y. G., Allanson, J., Cardone, P., Bonhomme, A., Carmona, J., Chatelle, C., Chennu, S., Conte, M., Dehaene, S., Finoia, P., Heinonen, G., Hersh, J. E., Kamau, E., Lawrence, P. K., Lupson, V. C., Meydan, A., Rohaut, B., Sanders, W. R., Sitt, J. D., ... Schiff, N. D. (2024). Cognitive motor dissociation in disorders of consciousness. *New England Journal of Medicine*, 391(7), 598–608.
<https://doi.org/10.1056/NEJMoa2400645>
- Briat, C., Gupta, A., & Khammash, M. (2016). Antithetic integral feedback ensures robust perfect adaptation in noisy biomolecular networks. *Cell Systems*, 2(1), 15–26. <https://doi.org/10.1016/j.cels.2016.01.004>

- Brown, E. N., Barbieri, R., Ventura, V., Kass, R. E., & Frank, L. M. (2002). The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2), 325–346. <https://doi.org/10.1162/08997660252741149>
- Cacioppo, J. T., & Berntson, G. G. (1994). Relationship between attitudes and evaluative space: A critical review, with emphasis on the separability of positive and negative substrates. *Psychological Bulletin*, 115(3), 401–423. <https://doi.org/10.1037/0033-2909.115.3.401>
- Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait–multimethod matrix. *Psychological Bulletin*, 56(2), 81–105. <https://doi.org/10.1037/h0046016>
- Carstensen, L. L. (2006). The influence of a sense of time on human development. *Science*, 312(5782), 1913–1915. <https://doi.org/10.1126/science.1127488>
- Carstensen, L. L., Isaacowitz, D. M., & Charles, S. T. (1999). Taking time seriously: A theory of socioemotional selectivity. *American Psychologist*, 54(3), 165–181. <https://doi.org/10.1037/0003-066X.54.3.165>
- Casali, A. G., Gosseries, O., Rosanova, M., Boly, M., Sarasso, S., Casali, K. R., Casarotto, S., Bruno, M.-A., Laureys, S., Tononi, G., & Massimini, M. (2013). A theoretically based index of consciousness independent of sensory processing and behavior. *Science Translational Medicine*, 5(198), 198ra105. <https://doi.org/10.1126/scitranslmed.3006294>
- Chatterjee, K., Chmelik, M., & Davies, J. (2016). A symbolic SAT-based algorithm for almost-sure reachability with small strategies in POMDPs. *Proceedings of the AAAI Conference on Artificial Intelligence*, 30(1), 3225–3232. <https://doi.org/10.1609/aaai.v30i1.10422>
- Chetrite, R., & Touchette, H. (2015). Nonequilibrium Markov processes conditioned on large deviations. *Annales Henri Poincaré*, 16, 2005–2057. <https://doi.org/10.1007/s00023-014-0375-8>
- Cox, D. R. (1972). Regression models and life-tables. *Journal of the Royal Statistical Society: Series B (Methodological)*, 34(2), 187–202. <https://doi.org/10.1111/j.2517-6161.1972.tb00899.x>
- Deakin, A., Mendl, M., Browne, W. J., Paul, E. S., & Hodge, J. J. L. (2018). State-dependent judgement bias in *Drosophila*: Evidence for evolutionarily primitive affective processes. *Biology Letters*, 14(2), Article 20170779. <https://doi.org/10.1098/rsbl.2017.0779>
- Diener, E., Lucas, R. E., & Scollon, C. N. (2006). Beyond the hedonic treadmill: Revising the adaptation theory of well-being. *American Psychologist*, 61(4), 305–314. <https://doi.org/10.1037/0003-066X.61.4.305>

- Doob, J. L. (1957). Conditional Brownian motion and the boundary limits of harmonic functions. *Bulletin de la Société Mathématique de France*, 85, 431–458. <https://doi.org/10.24033/bsmf.1494>
- Eldar, E., Rutledge, R. B., Dolan, R. J., & Niv, Y. (2016). Mood as representation of momentum. *Trends in Cognitive Sciences*, 20(1), 15–24. <https://doi.org/10.1016/j.tics.2015.07.010>
- Fredrickson, B. L., & Kahneman, D. (1993). Duration neglect in retrospective evaluations of affective episodes. *Journal of Personality and Social Psychology*, 65(1), 45–55. <https://doi.org/10.1037/0022-3514.65.1.45>
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. <https://doi.org/10.1198/016214506000001437>
- Haslbeck, J. M. B., Jover Martínez, A., Roefs, A. J., Fried, E. I., Lemmens, L. H. J. M., Groot, E., & Edelsbrunner, P. A. (2025). Comparing Likert and visual analogue scales in ecological momentary assessment. *Behavior Research Methods*, 57(8), Article 217. <https://doi.org/10.3758/s13428-025-02706-2>
- Hernán, M. A., & Robins, J. M. (2020). *Causal inference: What if*. Chapman & Hall/CRC.
- Kahneman, D., Wakker, P. P., & Sarin, R. (1997). Back to Bentham? Explorations of experienced utility. *The Quarterly Journal of Economics*, 112(2), 375–406. <https://doi.org/10.1162/003355397555235>
- Keramati, M., & Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife*, 3, e04811. <https://doi.org/10.7554/eLife.04811>
- Keren, H., Zheng, C., Jangraw, D. C., Chang, K., Vitale, A., Rutledge, R. B., Pereira, F., Nielson, D. M., & Stringaris, A. (2021). The temporal representation of experience in subjective mood. *eLife*, 10, e62051. <https://doi.org/10.7554/eLife.62051>
- Klasnja, P., Hekler, E. B., Shiffman, S., Boruvka, A., Almirall, D., Tewari, A., & Murphy, S. A. (2015). Microrandomized trials: An experimental design for developing just-in-time adaptive interventions. *Health Psychology*, 34(Suppl), 1220–1228. <https://doi.org/10.1037/hea0000305>
- Koob, G. F., & Le Moal, M. (2001). Drug addiction, dysregulation of reward, and allostasis. *Neuropsychopharmacology*, 24(2), 97–129. [https://doi.org/10.1016/S0893-133X\(00\)00195-0](https://doi.org/10.1016/S0893-133X(00)00195-0)
- Kotlarski, I. (1967). On characterizing the gamma and the normal distribution. *Pacific Journal of Mathematics*, 20(1), 69–76. <https://doi.org/10.2140/pjm.1967.20.69>

- Krantz, D. H., Luce, R. D., Suppes, P., & Tversky, A. (1971). *Foundations of measurement, Volume I: Additive and polynomial representations*. Academic Press.
- Lakens, D. (2017). Equivalence tests: A practical primer. *Social Psychological and Personality Science*, 8(4), 355–362. <https://doi.org/10.1177/1948550617697177>
- Landayan, D., Feldman, D. S., & Wolf, F. W. (2018). Satiation state-dependent dopaminergic control of foraging in *Drosophila*. *Scientific Reports*, 8, Article 5777. <https://doi.org/10.1038/s41598-018-24217-1>
- Langford, J. S., Hébert, E. T., Jones, D. R., Tonkin, S. S., Barker, B. A., Ulm, C., Shi, D., Becerra, J. A., & Businelle, M. (2026). Effects of ecological momentary assessment prompting schedule on affect measurement variability and associations with next-day health behaviors. *Assessment*. Advance online publication. <https://doi.org/10.1177/10731911261444953>
- Larsen, J. T., McGraw, A. P., & Cacioppo, J. T. (2001). Can people feel happy and sad at the same time? *Journal of Personality and Social Psychology*, 81(4), 684–696. <https://doi.org/10.1037/0022-3514.81.4.684>
- Lawlor, D. A., Tilling, K., & Davey Smith, G. (2016). Triangulation in aetiological epidemiology. *International Journal of Epidemiology*, 45(6), 1866–1886. <https://doi.org/10.1093/ije/dyw314>
- Lazic, S. E., Clarke-Williams, C. J., & Munafò, M. R. (2018). What exactly is ‘N’ in cell culture and animal experiments? *PLOS Biology*, 16(4), Article e2005282. <https://doi.org/10.1371/journal.pbio.2005282>
- Lipsitch, M., Tchetgen Tchetgen, E., & Cohen, T. (2010). Negative controls: A tool for detecting confounding and bias in observational studies. *Epidemiology*, 21(3), 383–388. <https://doi.org/10.1097/EDE.0b013e3181d61eeb>
- Little, R. J. A., & Rubin, D. B. (2019). *Statistical analysis with missing data* (3rd ed.). Wiley. <https://doi.org/10.1002/9781119482260>
- Luce, R. D., & Tukey, J. W. (1964). Simultaneous conjoint measurement: A new type of fundamental measurement. *Journal of Mathematical Psychology*, 1(1), 1–27. [https://doi.org/10.1016/0022-2496\(64\)90015-X](https://doi.org/10.1016/0022-2496(64)90015-X)
- Luhmann, M., Hofmann, W., Eid, M., & Lucas, R. E. (2012). Subjective well-being and adaptation to life events: A meta-analysis. *Journal of Personality and Social Psychology*, 102(3), 592–615. <https://doi.org/10.1037/a0025948>
- Manasova, D., Belloli, L. M. L., Rosenfelder, M. J., Willacker, L., Fló Rama, E., Valota, C., Hermann, B., Kaufmann, B. C., Pirastru, A., Derchi, C. C., Raiser, T., Valente, M., Sangare, A., Türker, B., Pyatigorskaya, N., Béranger, B., Colombo, M., Munoz-Musat, E., Escrichs, A., ... Sitt, J. D. (2026). Multimodal multicentre investigation of diagnostic and

- prognostic markers in disorders of consciousness. *Brain*, 149(4), 1381–1395. <https://doi.org/10.1093/brain/awaf412>
- Manski, C. F. (2003). *Partial identification of probability distributions*. Springer. <https://doi.org/10.1007/b97478>
- Meredith, W. (1993). Measurement invariance, factor analysis and factorial invariance. *Psychometrika*, 58, 525–543. <https://doi.org/10.1007/BF02294825>
- Munafò, M. R., & Davey Smith, G. (2018). Robust research needs many lines of evidence. *Nature*, 553(7689), 399–401. <https://doi.org/10.1038/d41586-018-01023-3>
- Parnia, S., Keshavarz Shirazi, T., Patel, J., Tran, L., Sinha, N., O'Neill, C., Roellke, E., Mengotto, A., Findlay, S., McBrine, M., Spiegel, R., Tarpey, T., Huppert, E., Jaffe, I., Gonzales, A. M., Xu, J., Koopman, E., Perkins, G. D., Vuylsteke, A., ... Deakin, C. D. (2023). AWAreness during RESuscitation—II: A multi-center study of consciousness and awareness in cardiac arrest. *Resuscitation*, 191, 109903. <https://doi.org/10.1016/j.resuscitation.2023.109903>
- Pearl, J., & Bareinboim, E. (2014). External validity: From do-calculus to transportability across populations. *Statistical Science*, 29, 579–595. <https://doi.org/10.1214/14-STS486>
- Percie du Sert, N., Hurst, V., Ahluwalia, A., Alam, S., Avey, M. T., Baker, M., Browne, W. J., Clark, A., Cuthill, I. C., Dirnagl, U., Emerson, M., Garner, P., Holgate, S. T., Howells, D. W., Karp, N. A., Lazic, S. E., Lidster, K., MacCallum, C. J., Macleod, M., Pearl, E. J., ... Würbel, H. (2020). The ARRIVE guidelines 2.0: Updated guidelines for reporting animal research. *PLOS Biology*, 18(7), Article e3000410. <https://doi.org/10.1371/journal.pbio.3000410>
- Petkus, A. J., Reynolds, C. A., Catts, V. S., Christensen, K., Finkel, D., Nygaard, M., Sachdev, P. S., Pedersen, N. L., & Gatz, M. (2025). Terminal increases in depressive symptoms in a multinational twin consortium. *Psychological Science*, 36(8), 637–655. <https://doi.org/10.1177/09567976251351022>
- Prochnow, T., Wang, W.-L., Wang, S., Li, J., Rothman, A. J., Intille, S. S., Hedeker, D., & Dunton, G. F. (2025). Understanding longitudinal ecological momentary assessment completion: Results from 12 months of burst sampling in the TIME study. *JMIR mHealth and uHealth*, 13, e67117. <https://doi.org/10.2196/67117>
- Protter, P. E. (2005). *Stochastic integration and differential equations* (2nd ed.). Springer. <https://doi.org/10.1007/978-3-662-10061-5>
- Røysland, K. (2011). A martingale approach to continuous-time marginal structural models. *Bernoulli*, 17(3), 895–915. <https://doi.org/10.3150/10-BEJ303>

- Rubin, D. B. (1976). Inference and missing data. *Biometrika*, 63, 581–592. <https://doi.org/10.1093/biomet/63.3.581>
- Scharfstein, D. O., Rotnitzky, A., & Robins, J. M. (1999). Adjusting for nonignorable drop-out using semiparametric nonresponse models. *Journal of the American Statistical Association*, 94(448), 1096–1120. <https://doi.org/10.1080/01621459.1999.10473862>
- Schuirmann, D. J. (1987). A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15, 657–680. <https://doi.org/10.1007/BF01068419>
- Self, S. G., & Liang, K.-Y. (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398), 605–610. <https://doi.org/10.1080/01621459.1987.10478472>
- Seow, H., Barbera, L., Sutradhar, R., Howell, D., Dudgeon, D., Atzema, C., Liu, Y., Husain, A., Sussman, J., & Earle, C. (2011). Trajectory of performance status and symptom scores for patients with cancer during the last six months of life. *Journal of Clinical Oncology*, 29(9), 1151–1158. <https://doi.org/10.1200/JCO.2010.30.7173>
- Solomon, R. L., & Corbit, J. D. (1974). An opponent-process theory of motivation: I. Temporal dynamics of affect. *Psychological Review*, 81(2), 119–145. <https://doi.org/10.1037/h0036128>
- Stone, A. A., Schwartz, J. E., Schkade, D., Schwarz, N., Krueger, A., & Kahneman, D. (2006). A population approach to the study of emotion: Diurnal rhythms of a working day examined with the Day Reconstruction Method. *Emotion*, 6(1), 139–149. <https://doi.org/10.1037/1528-3542.6.1.139>
- Vogel, N., Schilling, O. K., Wahl, H.-W., Beekman, A. T. F., & Penninx, B. W. J. H. (2013). Time-to-death-related change in positive and negative affect among older adults approaching the end of life. *Psychology and Aging*, 28(1), 128–141. <https://doi.org/10.1037/a0030471>
- Wang, J., Guo, J., Dove, A., Zhang, X., Yue, J., Bennett, D. A., & Xu, W. (2025). Trajectories of psychological and social well-being preceding death. *BMJ Mental Health*, 28(1), e301967. <https://doi.org/10.1136/bmjment-2025-301967>
- World Medical Association. (2024). WMA Declaration of Helsinki: Ethical principles for medical research involving human participants. <https://www.wma.net/policies-post/wma-declaration-of-helsinki/>
- Zrelec, V., Zini, M., Guarino, S., Mermoud, J., Oppliger, J., Valtat, A., Zeender, V., & Kawecki, T. J. (2013). *Drosophila* rely on learning while foraging under semi-natural conditions. *Ecology and Evolution*, 3(12), 4139–4148. <https://doi.org/10.1002/ece3.783>

About the Author

Stephen Takowsky has spent his life studying complex systems: how rules shape them, how people cooperate and compete within them, and how hidden constraints can change what becomes possible. His path has crossed finance, competitive strategy, professional esports, and the leadership of global online communities, ultimately leading to the Law of Fairness project, a decades-long effort to investigate whether fairness itself may reflect a deeper structure of conscious life.

Competing as “MrGameTheory,” Takowsky reached number one on the Civilization Revolution leaderboard and in the international Civilization IV and Civilization V leagues while pursuing a career in finance at Morgan Stanley in Beverly Hills. In 2016, aXiomatic, an ownership group led by Peter Guber and Ted Leonsis and backed by investors including Magic Johnson, Steve Case, and Tony Robbins, acquired a controlling interest in Team Liquid. Following the acquisition, Team Liquid selected Takowsky to captain its newly created Civilization team, later recognized by Guinness World Records as the first Civilization esports team. Across thousands of games, Civilization became an unconventional education in strategy: understand the whole system, think far beyond the next move, anticipate intelligent opposition, and keep searching when the obvious paths fail.

Strategy then became community leadership. Takowsky spent seven years at 505 Games, serving as Senior Digital Content & Engagement Manager and leading global communities for Terraria, which has surpassed 70 million copies sold, as well as Ghostrunner, WUCHANG: Fallen Feathers, Eiyuden Chronicle: Hundred Heroes, and other titles. He also founded the United Servers of Discord Alliance, bringing together leaders from major communities built around Roblox, Pokémon, Star Wars, League of Legends, Grand Theft Auto, Warframe, Animal Crossing, and other properties. Earlier, he served as administrator and producer for TF2Outpost, a major Steam digital-goods marketplace. Across these very different environments, he kept returning to the same insight: better structure can expand what people are capable of building together.

Throughout those years, one question kept following him: could the apparent unfairness of life conceal a deeper structure? That question became the Law of Fairness. Takowsky began developing the idea more than twenty years ago while studying Religious Studies and Philosophy/Law and Society at the University of California, Riverside, and gave it its first public form in his 2009

book, *Of Grandeur*. In the years that followed, he sought out academics and serious thinkers across many disciplines, asking for insights, criticism, and sources that could help separate fact from speculation. What began as a philosophical intuition gradually became a research program built to face mathematics, measurement, competing explanations, and evidence.

The ten-book Law of Fairness series follows that question outward, from consciousness and scientific testing into ordinary life, meaning, and civilization. If the Law is true, then there may be no “free happiness” in the sense of positive experience remaining unresolved by the lifetime balance the hypothesis predicts. That possibility raises a profound practical question: what if humanity has been pursuing the wrong model of utopia? Rather than organizing life around the endless pursuit of pleasure, the series explores whether greater self-moderation, less needless suffering, and better-designed institutions could help remove some of the forces that keep people and societies trapped in conflict, corruption, and instability.

Takowsky’s fifty-year horizon is designed so that building a better future does not require tearing apart the present. It gives institutions and communities time to test reforms, build infrastructure, prepare future leaders, improve technology, and correct mistakes before they spread. Just as importantly, it gives today’s generation the chance to be remembered for beginning something larger than itself: laying the foundations for a more transparent, peaceful, and prosperous world it may never live to see.

The project’s future is collaborative. Takowsky welcomes researchers, universities, laboratories, funders, technologists, students, critics, writers, artists, community builders, and volunteers willing to bring rigor, imagination, and courage to one of humanity’s oldest questions. The next breakthrough may come from a theorem, an instrument, a dataset, a counterexample, or an experiment no one has yet imagined. *The Law of Fairness* began as one person’s question. Its next chapter belongs to everyone willing to help discover what is true and build what that knowledge makes possible.

Explore the project and its official communities at LawOfFairness.com. Research, scholarly, media, funding, volunteer, and collaboration inquiries: team@lawoffairness.com.

